

Zara Sales Analysis & EDA

Project Overview

This project focuses on performing Exploratory Data Analysis (EDA) on Zara sales data to understand customer buying behavior, product performance, pricing strategy, promotion effectiveness, and seasonal trends.

The main goal of this analysis is to generate business insights that can help improve sales performance and marketing decisions.

Objectives

- Analyze top-selling products
- Understand the impact of promotions on sales
- Compare category-wise and brand-wise performance
- Analyze seasonal vs non-seasonal products
- Study pricing impact on sales volume
- Identify how product placement affects sales
- Generate business recommendations from data

Dataset Information

The dataset contains product-related sales information such as:

- Product ID
- Product Position
- Promotion Status
- Product Category
- Seasonal Information
- Sales Volume
- Brand
- Product Name & Description
- Price
- Currency
- Store Section

Tools & Libraries Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

Data Cleaning

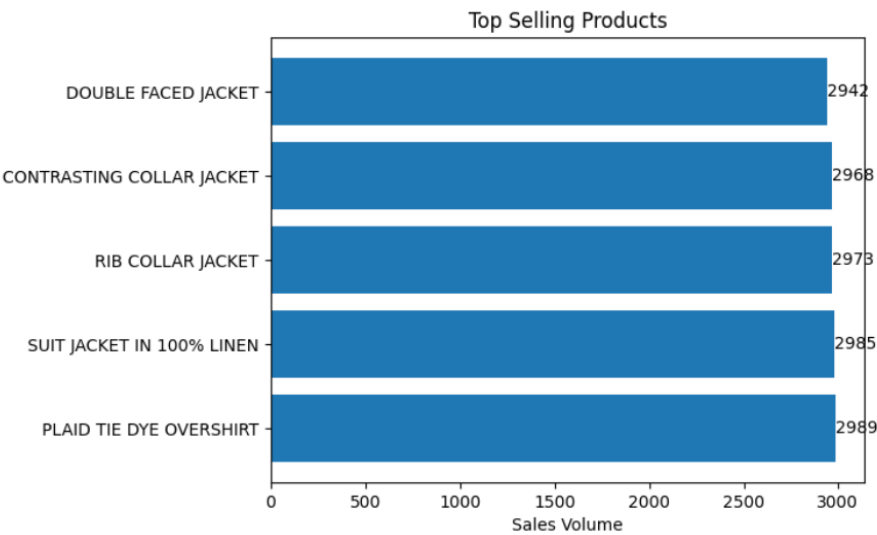
The following preprocessing steps were performed:

- Checked missing values
- Removed duplicates
- Handled null values
- Created new calculated columns
- Converted data into analysis-friendly format

Exploratory Data Analysis (EDA)

1. Top Selling Products

Analyzed products with the highest sales volume.



Insight

Key Insights

- 1. Products like **PLAID TIE-DYE OVERSHIRT**, **SUIT JACKET IN 100% LINEN**, and **RIB COLLAR JACKET** are among the highest-selling items.
- 2. Jacket and overshirt products dominate customer demand, showing strong preference for outerwear and fashion essentials.
- 3. Sales volume among top products is very close, meaning multiple products are consistently performing well instead of relying on only one bestseller.

Business Impact

1. Inventory Planning

These products should always remain in stock. Stock-outs on these items may directly reduce revenue opportunities.

Recommendation:

- Maintain higher inventory levels for top-performing products
- Monitor stock availability weekly

2. Marketing Focus

Best-selling products can be used in:

- Homepage banners
- Social media ads
- Email campaigns
- Seasonal promotions

Recommendation:

- Prioritize these products in promotional campaigns.
- Highlight them as “Best Sellers” on website/store

3. Product Strategy

Customers show a strong interest in jackets and premium fashion wear.

Recommendation:

- Expand similar product styles and collections
- Launch related variants (colors, fits, seasonal editions)

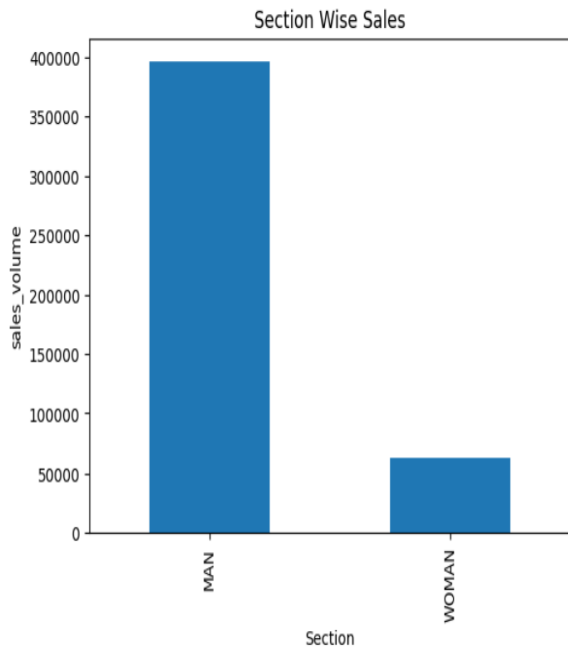
4. Revenue Optimization

Since these products already have strong demand, even small pricing optimization can improve overall revenue.

Recommendation:

- Test slight premium pricing on high-demand products
- Bundle top-selling products with accessories.

2. Section-wise Sales Analysis



Key Insights

The **MAN section dominates overall sales volume** compared to the WOMAN section. Sales from the MAN category are significantly higher, showing stronger customer demand and better product performance. The WOMAN section contributes a much smaller share of total sales, indicating potential growth opportunities.

Business Impact

1. Strong Demand in Men's Fashion

Men's products are currently the main revenue-driving segment.

Recommendation:

- Continue expanding high-performing men's collections
- Increase inventory for best-selling men's products
- Launch more variations of successful items

2. Improvement Opportunity in Women's Section

Lower sales in the WOMAN section may indicate the following:

- Limited product variety
- Pricing mismatch
- Weak promotions
- Less attractive product positioning

Recommendation:

- Improve women's product assortment
- Introduce trend-based collections
- Run targeted promotions for women customers.

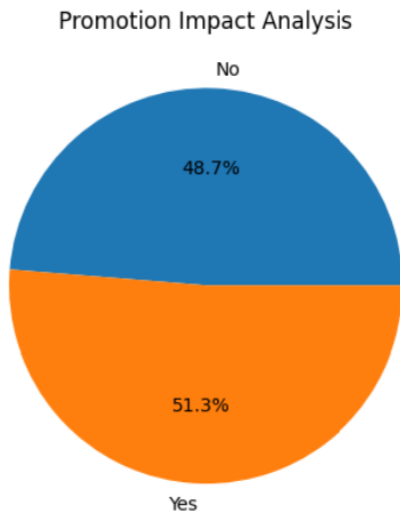
3. Marketing Strategy Optimization

The business can use MEN products for stable revenue while improving WOMAN category growth.

Recommendation:

- Create separate marketing campaigns for WOMAN section
- Use influencers/social media campaigns targeting female audience
- Promote seasonal women's collections more aggressively

3. Promotion Impact Analysis



Key Insights

Promoted products account for slightly higher sales performance compared to non-promoted products. Around **51.3%** of sales are influenced by promotional activities, while **48.7%** come from non-promoted products. The difference is not extremely large, which shows that Zara products already have strong organic demand even without heavy discounts.

Business Impact

1. Promotions Positively Influence Sales

Promotional campaigns help increase customer purchases and product visibility.

Recommendation:

- Continue running smart promotional campaigns
- Use promotions during seasonal launches and clearance periods

2. Strong Brand Value

Nearly half of the sales come without promotions, showing strong brand loyalty and customer trust.

Recommendation:

- Maintain premium brand positioning
- Avoid excessive discounting that may reduce perceived brand value

3. Opportunity for Strategic Promotions

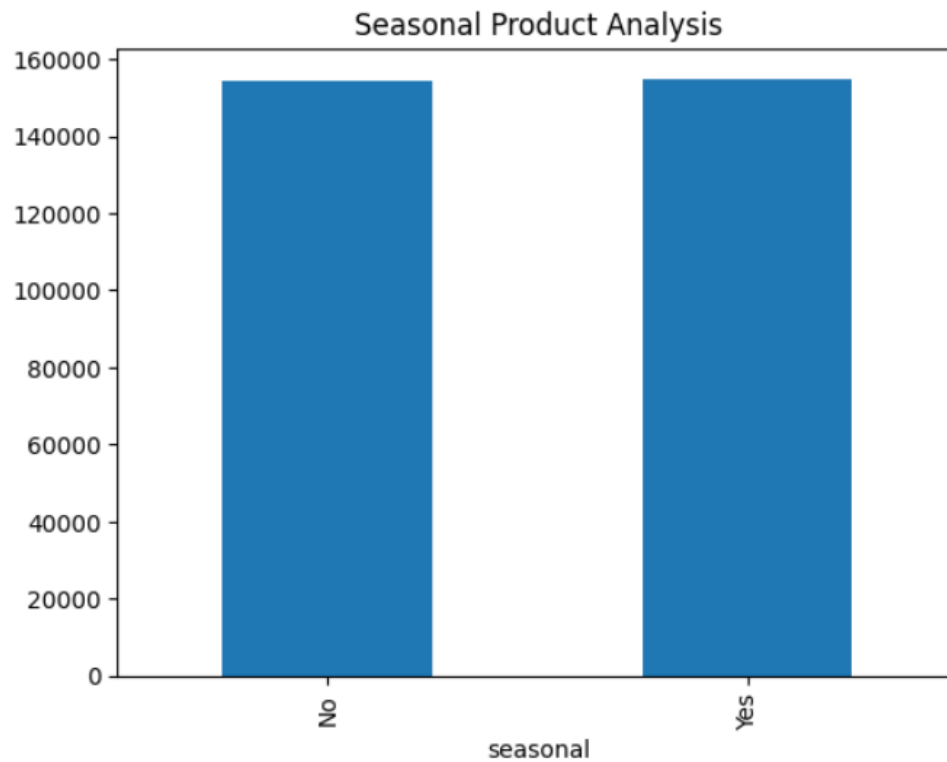
Promotions should focus on:

- Medium-performing products
- Seasonal inventory
- New product launches

Recommendation:

- Use limited time offers
- Bundle products to increase average order value
- Promote slow-moving inventory strategically

4. Seasonal Product Analysis



Key Insights

Seasonal and non-seasonal products show very similar sales performance. Seasonal products have a slightly higher sales volume, indicating that season-based demand positively influences customer purchases. The small difference suggests that Zara maintains stable demand throughout the year, not only during seasonal periods.

Business Impact

1. Seasonal Products Still Matter

Seasonal collections help boost customer engagement and trend-based purchases.

Recommendation:

- Launch seasonal collections before peak demand periods
- Focus on trend-driven seasonal marketing campaigns

2. Stable Non-Seasonal Demand

Non-seasonal products continue generating strong sales consistently.

Recommendation:

- Maintain core evergreen products throughout the year
- Keep best-selling essentials permanently available

3. Balanced Product Portfolio

Zara benefits from having both:

- Trend-driven seasonal products
- Stable year-round products

Recommendation:

- Balance inventory between seasonal and evergreen collections
- Use seasonal launches to attract traffic while evergreen products maintain consistent revenue

5. Price vs Sales Relationship



Key Insights

Most products are concentrated in the low to medium price range, showing Zara's strong focus on affordable fashion. Lower-priced products generally achieve higher sales volume because they are accessible to a larger customer base. Some high-priced products still generate strong sales, indicating that premium products also have demand among customers. Extremely expensive products have lower sales volume, which is expected due to a smaller target audience.

Business Impact

1. Affordable Products Drive Customer Traffic

Mid-range and affordable products are the main contributors to sales volume.

Recommendation:

- Continue offering competitively priced fashion products
- Use affordable products to attract more customers

2. Premium Products Increase Revenue

High-priced products may sell less frequently but can generate strong profit margins.

Recommendation:

- Promote premium collections strategically
- Highlight premium products in featured sections and campaigns

3. Balanced Pricing Strategy is important.

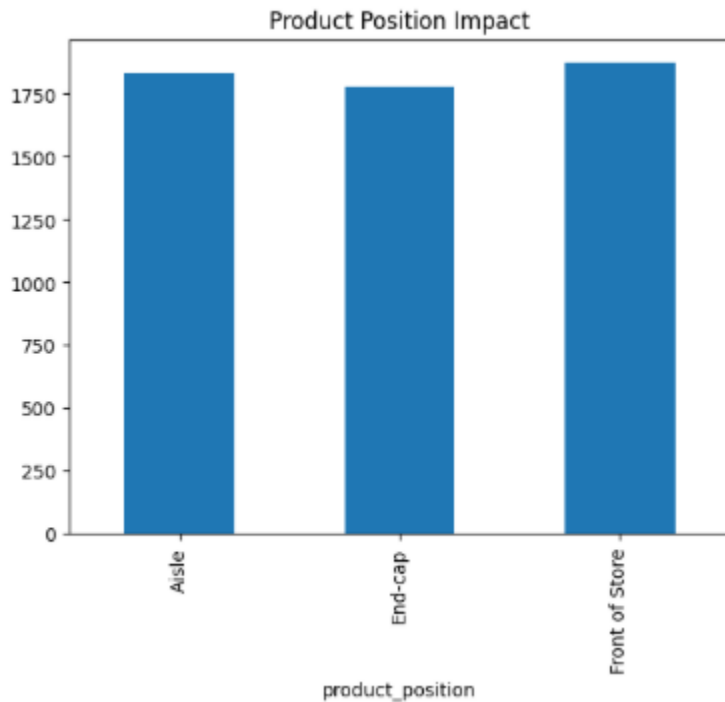
The business benefits from maintaining both:

- High-volume affordable products
- High-margin premium products

Recommendation:

- Maintain a balanced product portfolio
- Use affordable items for traffic generation and premium items for revenue growth

7. Product Position Analysis



Key Insights

Products placed at the front of the store generated the highest average sales volume. **Aisle placements** also performed strongly, showing that product visibility plays a major role in customer buying behavior. **End-cap positions** generated slightly lower sales compared to other positions but still maintained stable performance.

Business Impact

1. Store Visibility Directly Impacts Sales

Products placed in high-visibility areas attract more customer attention and increase purchase chances.

Recommendation:

- Place best-selling and premium products at the Front of Store
- Use eye-catching displays for featured products

2. Strategic Product Placement Can Increase Revenue

Proper store layout can improve customer engagement and product discovery.

Recommendation:

- Rotate promotional and seasonal products into high-traffic locations
- Use aisle displays for trending items

3. Optimize Underperforming Areas

End-cap placement shows slightly lower performance compared to Front of Store.

Recommendation:

- Improve end-cap visual merchandising
- Add promotional signage or discounts to boost attention

8. Correlation Heatmap



1. Price vs Revenue → Strong Positive Correlation (0.75)

Higher-priced products are generating significantly higher revenue. Premium products contribute strongly to business earnings. Zara can focus more on premium collections and high-value fashion items.

Business Recommendation:

- Increase visibility of premium products
- Use premium product marketing campaigns
- Maintain strong inventory for high-priced best sellers

2. Sales Volume vs Revenue → Moderate Positive Correlation (0.50)

Products with higher sales volume also help increase revenue. Volume-based products are important for overall business growth.

Business Recommendation:

- Keep fast-selling products always in stock.
- Use promotions on high-demand products
- Improve inventory planning for popular items

3. Price vs Sales Volume → Slight Negative Correlation (-0.071)

As product prices increase, sales volume slightly decreases. Customers tend to purchase lower-priced products more frequently.

Business Recommendation:

- Maintain a balance between affordable and premium products
- Use affordable products to attract customers
- Upsell customers toward premium products